PROPRIETARY & COMPETITION SENSITIVE (DFARS 252.227-7018) | Nontraditional Defense Contractor

EXECUTIVE PROPOSAL BRIEF | SOVEREIGN PRIME CONTRACTOR

SOLICITATION IDENTIFIER: DIU-AOI-2026-AUTONOMY
TARGET AGENCY: Defense Innovation Unit (DIU) / Office of the Under Secretary of Defense (R&E)
PROGRAM TITLE: Contested Logistics & Edge Autonomy Resiliency
ACQUISITION VEHICLE: 10 U.S.C. § 4022 Commercial Solutions Opening (Prototype Other Transaction)
TOTAL PROTOTYPE ALLOCATION: $1,650,000.00 USD (100% Solo-Prime Allocation)
SUBMISSION FORMAT: 5-Page Commercial Solution Brief

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1. OPERATIONAL ENTITY & PRIME AUTHORITY

- **Prime Contractor:** HVF Omni-Industrial Matrix (100% Unencumbered Prime)
- **Corporate Identifiers:** CAGE: 1AHA8 | UEI: S1M4ENLHTDH5 | Small Business Concern
- **Contracting Status:** Nontraditional Defense Contractor pursuant to 10 U.S.C. § 4022(d)(1)(A)
- **Primary SME & Principal Investigator:** Jeffery Humphrey, Founder & CEO (Apex Architect)
- **Headquarters & Prototyping Laboratory:** Oklahoma City, Oklahoma, USA
- **Primary Operational Contact:** humphreyvirtualfarm@gmail.com

2. ARCHITECTURAL SOLUTION: PROJECT EBONY

Project Ebony delivers sovereign, zero-cloud autonomous resilience for tactical logistics, cyber-physical SCADA, and distributed uncrewed platforms in heavily contested Anti-Access/Area Denial (A2/AD) and electronic warfare (EW) environments. Operating entirely on bare-metal silicon with zero external cloud dependencies, Project Ebony decouples physical actuation, cognitive anomaly detection, and cryptographic provenance across an air-gapped **Three-Brain Architecture**:

- **Brain One (Deterministic Controller & Kinetic Guillotine):** Executes hard-real-time SCADA and kinetic actuation logic on bare-metal ARM Cortex-M7 and RISC-V microcontrollers. Features the 'Kinetic Guillotine'—a hardware-anchored physical interlock that severs actuator power circuits in less than 1.0 microsecond upon boundary threshold breaches (voltage, current, pressure, vibration). Compromised software cannot override an open physical circuit.
- **Brain Two (Offline Cognitive Analyst):** Executes edge neural inference locally on dedicated acceleration silicon, identifying multi-spectral anomalies and phenotypic drift in under 5.0ms without RF leakage or cloud queries.
- **Brain Three (Local Merkle Arbiter):** Maintains an immutable local Merkle DAG ledger logging every state transition in sub-millisecond cycles (<0.2ms) satisfying NIST SP 800-230 integrity standards with zero external dependencies.

3. PERFORMANCE-BASED MILESTONE SCHEDULE ($150,000 MOBILIZATION)

HVF Omni-Industrial Matrix executes under performance-based payable milestones incorporating a front-loaded mobilization tranche (M1A) to fund long-lead hardware procurement, micro-architecture bench fabrication, and testbed staging within 15–30 days of award:

Milestone	Window	Deliverable Focus & Verification Criteria	Allocation
M1A: Mobilization	**Days 15–30**	Systems Engineering Plan (SEP), Post-Award Kickoff, Long-Lead Silicon Procurement Plan, and Prototyping Lab Initialization.	**$150,000.00**
M1B: HW Integration	Month 3	Bare-metal deployment of Brain One & Three; sub-0.15ms SHA-256 Merkle DAG state validation on ARM/RISC-V hardware.	$200,000.00
M2: Hardening	Month 6	Hardware-in-the-Loop (HIL) stress-testing across 10,000 fault-injection cycles; <1.0us Kinetic Guillotine circuit cutoff verification.	$450,000.00
M3: AI Optimization	Month 9	Brain Two offline neural inference deployment; multi-spectral sensor spoofing detection (<5.0ms) with zero outbound RF emissions.	$450,000.00
M4: Field Demo	Month 12	48-hour continuous air-gapped operational field demonstration on live pumping testbed under simulated EW denial; TRL-7 data package.	$400,000.00
TOTAL PROTOTYPE CONTRACT EFFORT (10 U.S.C. § 4022)			**$1,650,000.00**

4. INTELLECTUAL PROPERTY & DATA RIGHTS (DFARS 252.227-7018)

HVF Omni-Industrial Matrix asserts 100% exclusive, unencumbered ownership of all Background Intellectual Property, bare-metal source code, hardware schematics, firmware architectures, and neural models developed prior to or outside of this agreement pursuant to DFARS 252.227-7018. The Government receives negotiated commercial prototype evaluation rights strictly for testing and evaluation during the Prototype OT. HVF retains absolute commercial title, sole licensing authority, and exclusive follow-on production rights under 10 U.S.C. § 4022(f).

SUBMITTED ON BEHALF OF PRIME CONTRACTOR:
Jeffery Humphrey, Founder & Chief Executive Officer
Apex Architect & Principal Investigator | HVF Omni-Industrial Matrix
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